Nicholas Young

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PROFESSIONAL SUMMARY

- Hardware verification engineer with 5+ years of experience verifying SoC designs at Intel in a pre-silicon environment
- Familiarity with industry standard verification tools such as SystemVerilog and UVM for testbench development, and the infrastructure surrounding them
- Verified the functionality of fabrics within the SoC which allow all parts of the chip to communicate with each other
- Trusted to take on two tours of duty within Intel, working with the front end tools team and an international IP integration verification team for short periods of time
- Explored applying generative AI to SoC verification to understand where and how best the tools can be used

TECHNICAL SKILLS

- **Programming Languages:** SystemVerilog, Python, C/C++
- **Software Development Tools** - Linux, CLI, Git
- **Verification:** Testbench Development, UVM, Coverage, Digital Simulation, Waveform Debug
- **Artificial Intelligence** - GitHub Copilot Chat, LLM-assisted scripting and debug, Agentic workflows

PROFESSIONAL EXPERIENCE

Intel - Pre-Silicon Hardware Verification Engineer — Hillsboro, OR
SoC Chassis Verification Team — June 2021 - Present

- Verified the design of the fabrics at the SoC level, allowing all IPs within the SoC to communicate with each other
- Worked closely with the design and architecture teams to prepare test plans and coverage plans, allowing thorough verification of the fabrics in a pre-silicon environment
- Developed SystemVerilog testbenches using UVM, enabling proper simulation and verification of the SoC with various combinations of IPs and Subsystems enabled
- Programmed tests, monitors, and checkers to exercise all of the fabric's functionality and determine whether it was acting correctly according to the design specifications
- Debugged critical connection issues blocking SoC simulation bring up, enabling other teams to begin integrating IPs into the SoC
- Managed regressions comprising thousands of tests a week for the fabrics team, doing initial triage of failures and assigning them to team members accordingly in order to catch any issues that were uncovered
- Composed python scripts to help simplify and accelerate my team's normal workflow and increase productivity

Front End Tools Team — May 2022 - December 2022

- Volunteered to take a temporary position working to improve the tools surrounding the pre-silicon simulation environment
- Developed scripts and tools to ensure the health of the validation codebase and enforce coding standards within it
- Granted administrator privileges to control updates to the validation codebase to ensure smooth operations for the rest of the team

IP Integration Verification Team — March 2023 - September 2023

- Selected to work with an international team on a six month assignment to enhance cross team collaboration
- Worked independently to validate the IP due to being the only team member in the United States time zones
- Uncovered a major bug missed by previous teams, allowing it to be fixed just before the chip was being made in silicon resulting in millions of dollars in savings for Intel

EDUCATION

University of Michigan Ann Arbor, MI — April 2020, April 2021

- Master of Science in Engineering in Electrical and Computer Engineering, Embedded Systems — GPA: 3.75
- Bachelor of Science in Engineering in Computer Engineering, Summa Cum Laude — GPA: 3.78
- International Engineering Minor